**8** A tennis ball with momentum 2.3 kg m / s is struck by a tennis racket.

The racket causes the direction of motion of the ball to reverse.

The racket exerts an average force of 1500 N during the impact with the ball. The racket and ball are in contact with each other for a time of 0.0024 s.

What is the magnitude of the momentum of the ball as it leaves the tennis racket?

**A** 1.3 kg m / s **B** 3.6 kg m / s **C** 5.9 kg m / s **D** 8.6 kg m / s